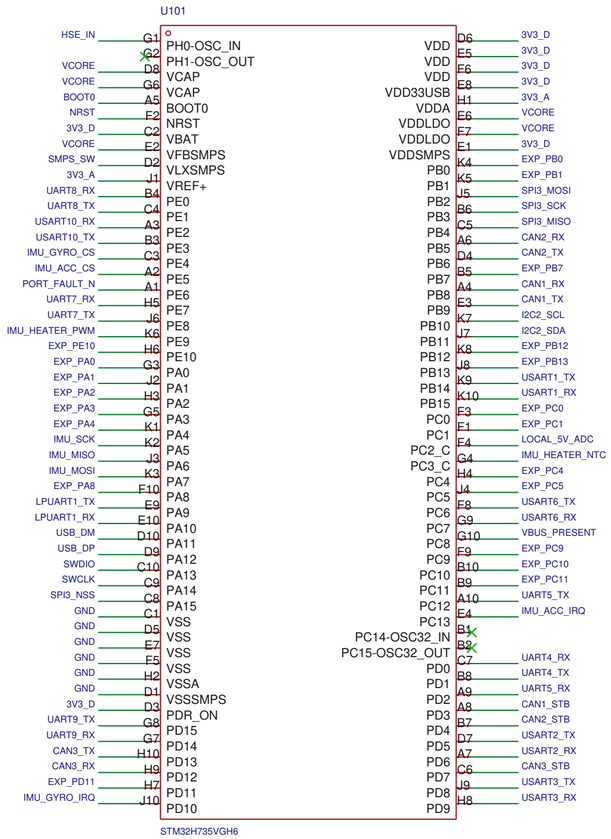
C01A_H735主控与信号分配

480 MHz / 8 MHz HSE BYPASS / STM32 + FreeRTOS

核心板：3 CAN + 4 UART
SPI1：BMI088，双片选
SPI3 / I2C2：独立外设接口
USB FS：自供电设备

上层扩展板：7 组其余 UART
板间 J801 / J802：2 x 30 pin
PA0 / PA1 / PB0 / PB1：自定义排针
PC8：USB VBUS 检测
PE6：5V_PORT 故障反馈

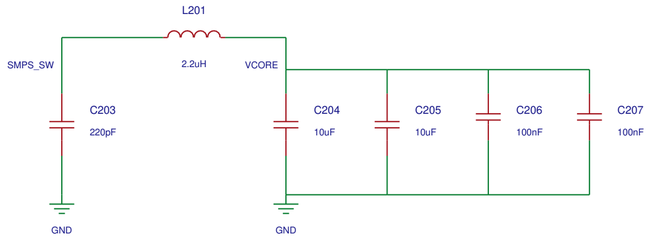
与旧固件不同：IMU SPI4 → SPI1
ACC_CS → PE5；VBUS → PC8
PA9 / PA10 用于 LPUART1
PE2 / PE3 用于 USART10
旧引脚复用配置不可直接沿用



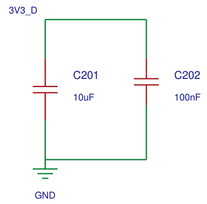
Native-source geometry preview / not an editor-rendered or fabrication drawing

C01B_主控SMPS与电源去耦

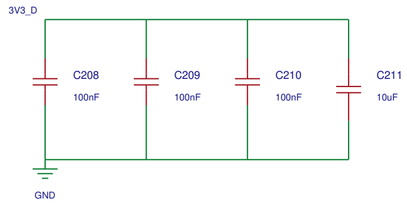
内部 SMPS / VCORE



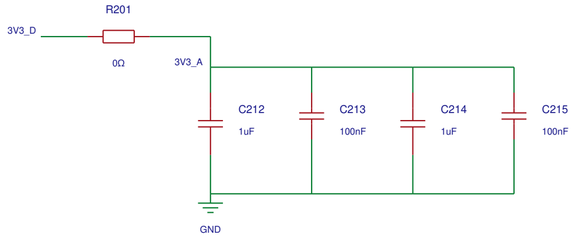
SMPS 输入



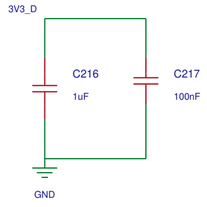
数字电源



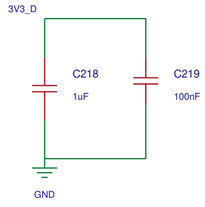
模拟电源 / VREF



USB 电源



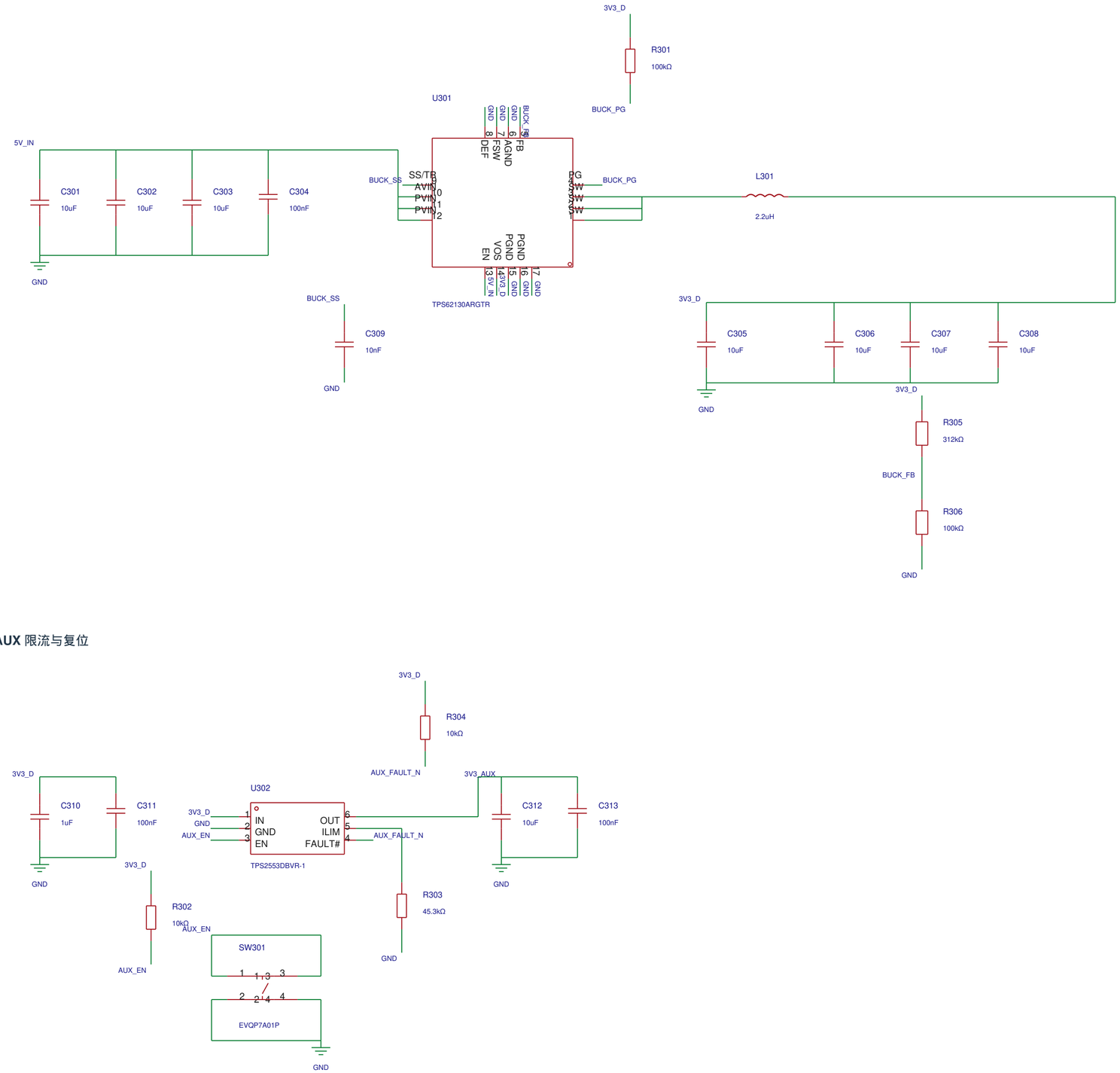
VBAT 电源



Native-source geometry preview / not an editor-rendered or fabrication drawing

C02_3V3降压与扩展电源保护

5V → 3.296V / TPS62130A

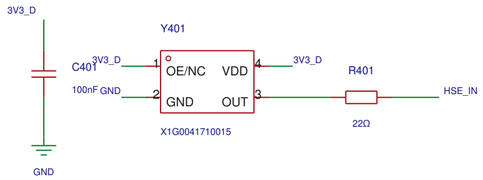


3V3_AUX 限流与复位

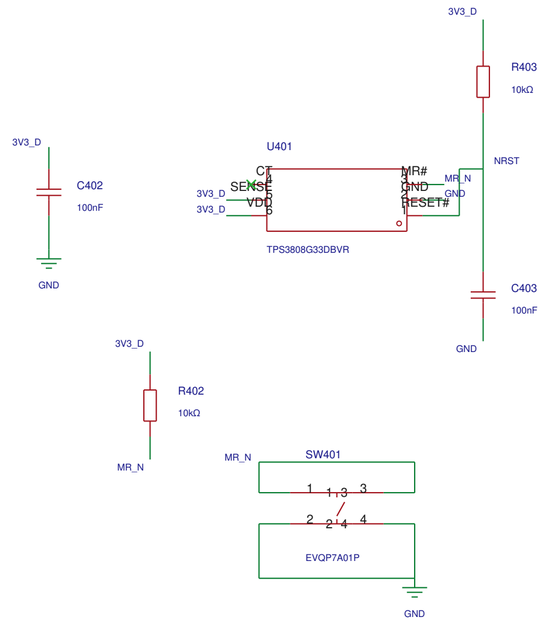
Native-source geometry preview / not an editor-rendered or fabrication drawing

C03_时钟复位启动与SWD

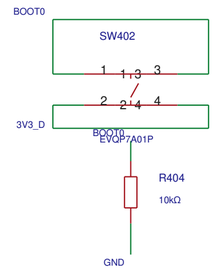
8 MHz HSE BYPASS



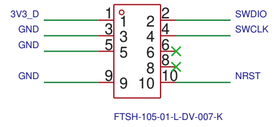
复位监控 / 手动复位



BOOT0 默认下拉



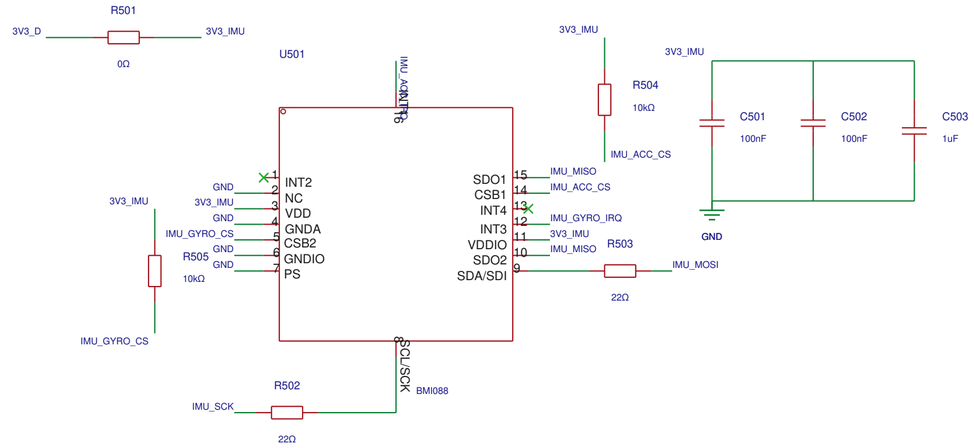
SWD / 不由调试器供电



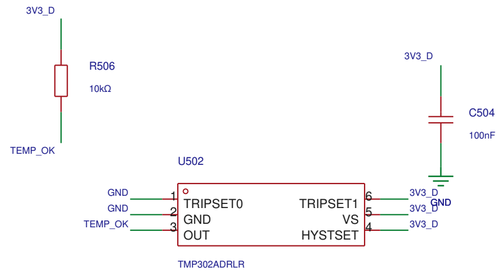
Native-source geometry preview / not an editor-rendered or fabrication drawing

C04_BMI088与加热安全控制

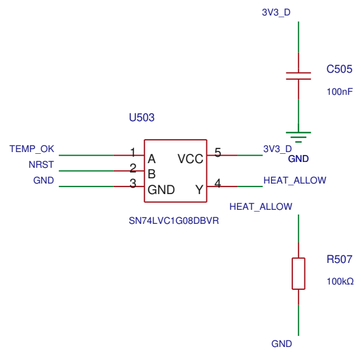
BMI088 / SPI 与电源



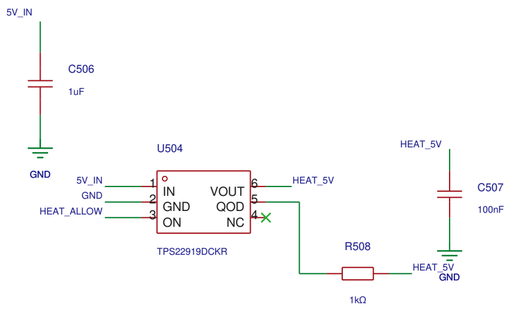
60°C 硬件温控



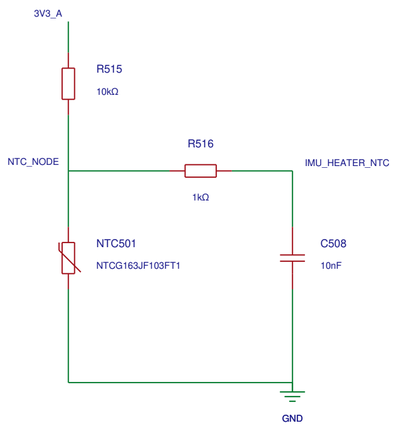
NRST 安全门



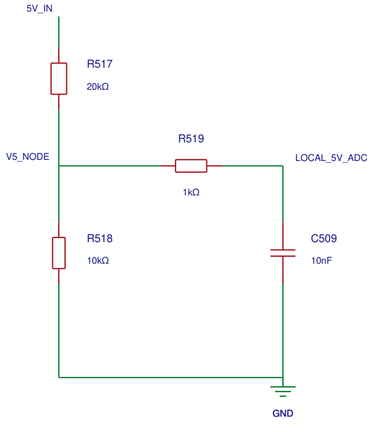
加热电源开关



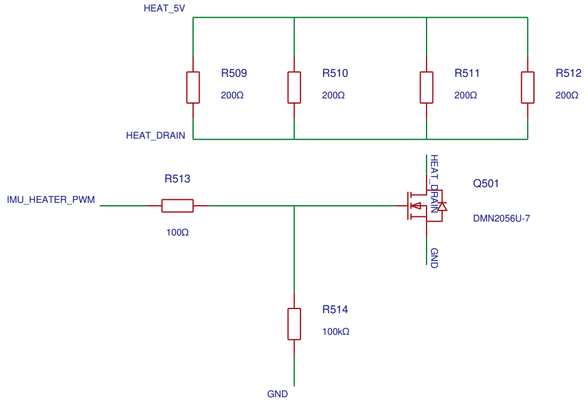
IMU 温度 ADC



5V 输入 ADC



加热电阻 / 低边 PWM

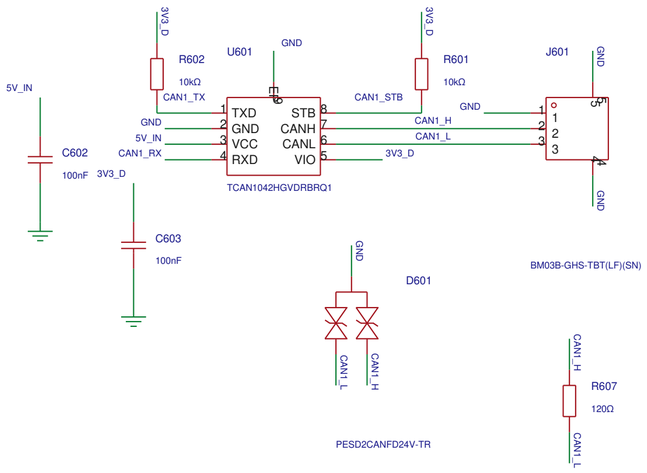


IMU 热隔离岛 / PCB 设计约束
岛内：BMI088、就近去耦、4×200Ω加热电阻、NTC、TMP302
岛外：MCU、两级降压、电感、加热开关和功率 MOSFET
岛边开槽并留机械桥；电源/地/信号回流成组跨桥
隔热同时保留刚性，不能把岛做成易共振的悬臂
最终槽宽、桥宽、岛尺寸需热仿真/振动与样机验证
恒温目标先按45°C验证；60°C硬件禁止不是正常设定点
SPI1 PA5/PA6/PA7，ACC_CS=PE5；旧SPI4固件需修改

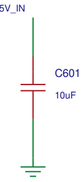
Native-source geometry preview / not an editor-rendered or fabrication drawing

C05_CAN三路收发与保护

CAN1 / GH1.25 直立 / 端点电阻 DNP

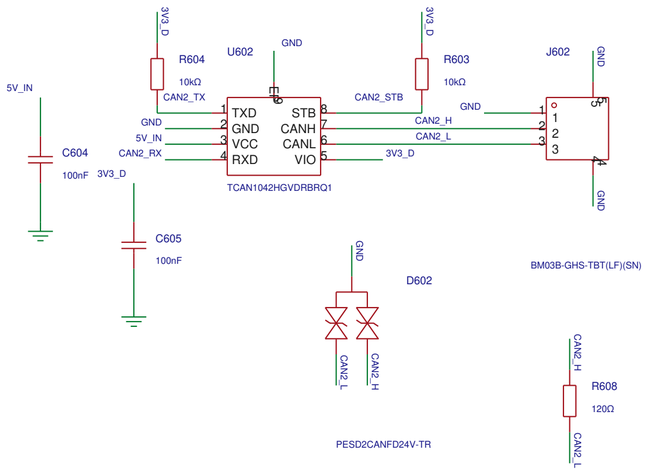


120Ω 默认不装
仅总线端点装配



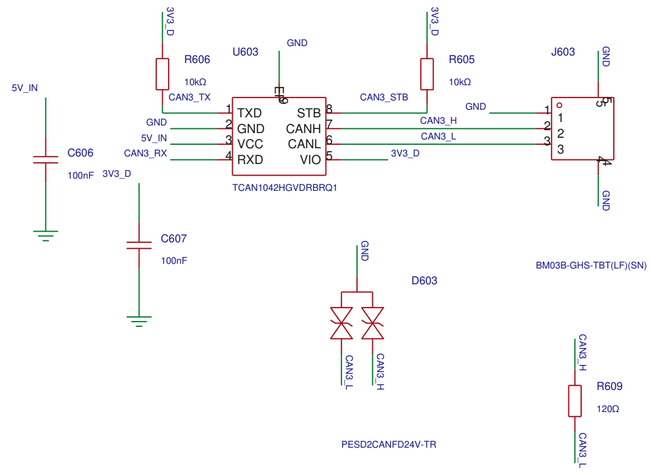
GH 针序：1 GND / 2 CANH / 3 CANL；固定焊脚接地
每条总线全网只装两只 120Ω；扩展板端接不得重复
CAN 不携带 48V；STB 复位默认上拉进入待机

CAN2 / GH1.25 直立 / 端点电阻 DNP



120Ω 默认不装
仅总线端点装配

CAN3 / GH1.25 直立 / 端点电阻 DNP

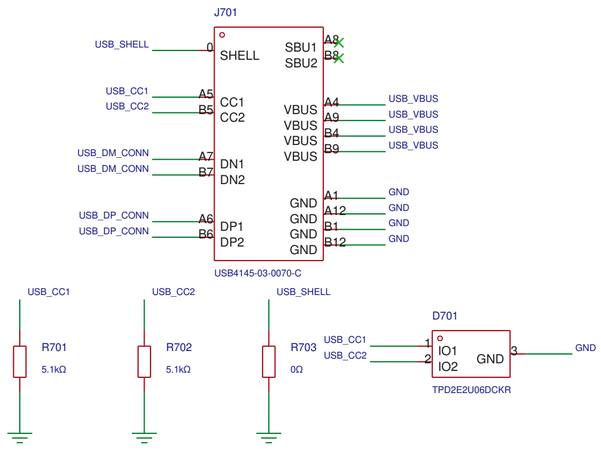


120Ω 默认不装
仅总线端点装配

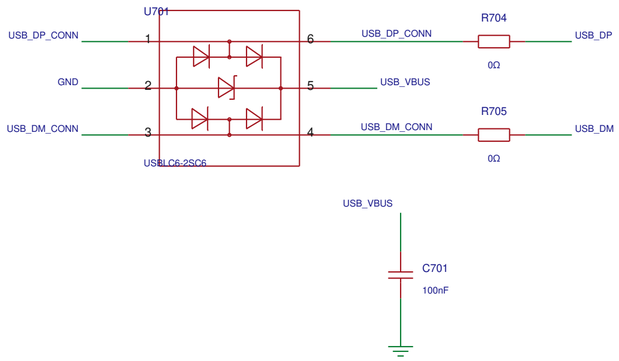
Native-source geometry preview / not an editor-rendered or fabrication drawing

C06_USB设备接口与VBUS检测

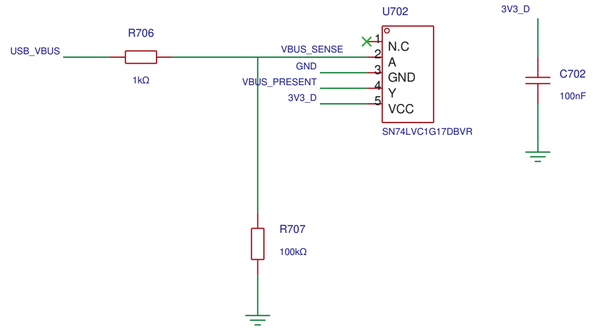
USB4145 / 直立 / 设备接口



USB2.0 ESD / 串联预留



VBUS_PRESENT → PC8 GPIO

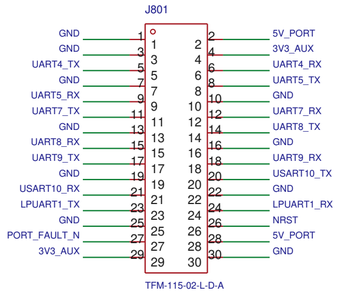


USB VBUS 仅检测，不给核心板供电；CC1 / CC2 各 5.1k 下拉
四个外壳焊脚同为 pad 0，属于一个 USB_SHELL 网络；A8 / B8 不连接
上层扩展板须避让 7.46 mm 插座、线缆插头和拔插操作空间

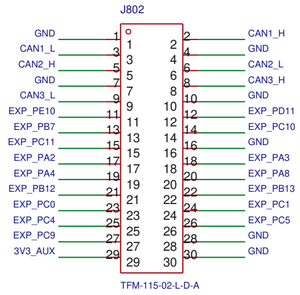
Native-source geometry preview / not an editor-rendered or fabrication drawing

C07_板间连接器与测试点

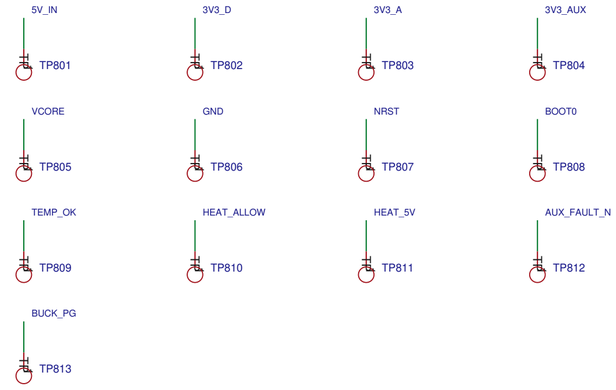
J801 / 7 组串口 / 电源与复位



J802 / CAN / 其余可复用 GPIO



测试点 / 无钢网开口

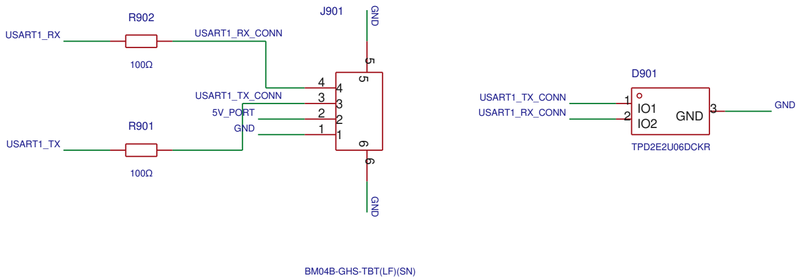


核心板在下：TOP = TFM-115-02-L-D-A
扩展板在上：BOTTOM = SFM-115-02-L-D-A-K-TR
两组 30pin；装配方向与旧版相反；禁止沿用旧 50pin 针序
5V_PORT 与 3V3_AUX 均为受限流保护的输出，不接外部电源

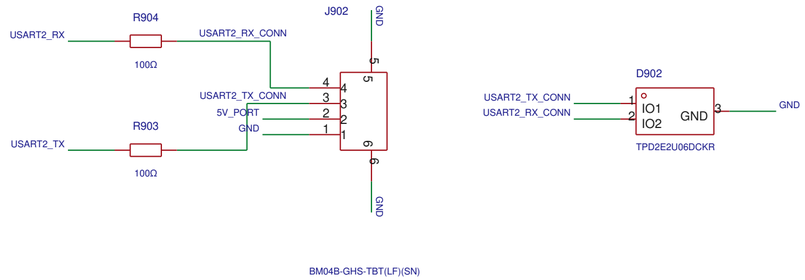
Native-source geometry preview / not an editor-rendered or fabrication drawing

C08_四组UART直立接口

USART1 / 3.3V UART / GH1.25 直立



USART2 / 3.3V UART / GH1.25 直立



USART3 / 3.3V UART / GH1.25 直立

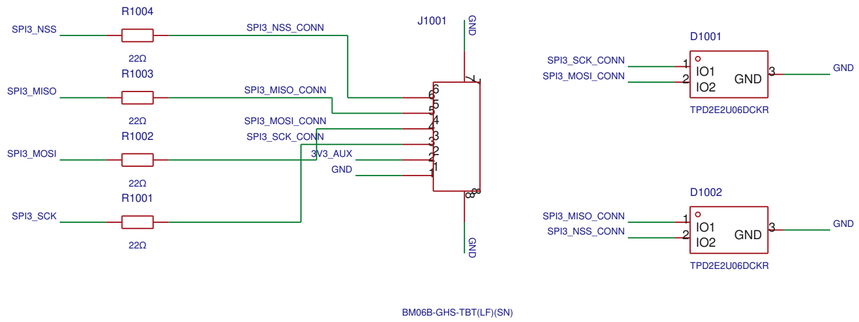
USART6 / 3.3V UART / GH1.25 直立

四针统一：1 GND / 2 +5V_PORT / 3 MCU_TX / 4 MCU_RX；外部设备 TX/RX 交叉接线
5V_PORT 由所有核心 UART 与扩展板共同分配，连续负载预算 400mA，总限流详见 C09
UART4/5/7/8/9、USART10、LPUART1 从 J801 送往上层扩展板

Native-source geometry preview / not an editor-rendered or fabrication drawing

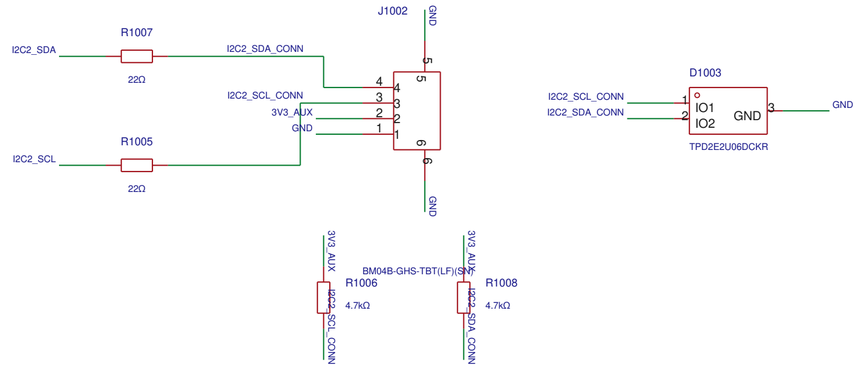
C09_SPI_I2C自定义排针与接口供电

SPI3 / 3.3V / 六针 GH



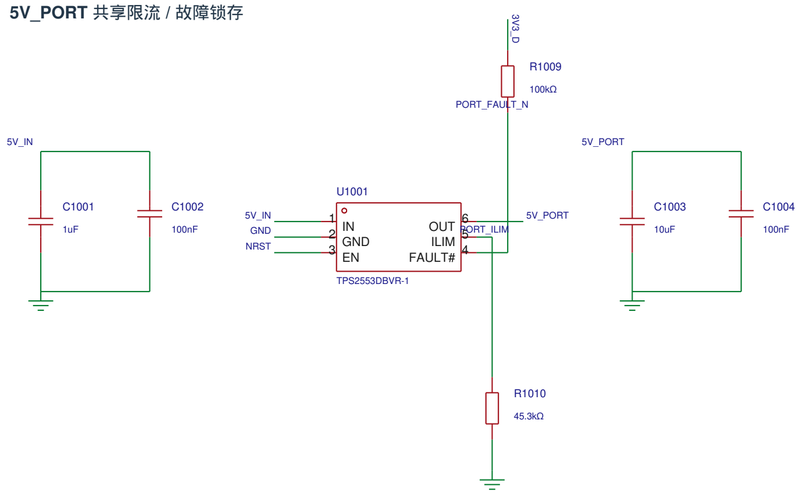
1 GND / 2 3V3_AUX / 3 SCK
4 MOSI / 5 MISO / 6 NSS

I2C2 / 3.3V / 四针 GH



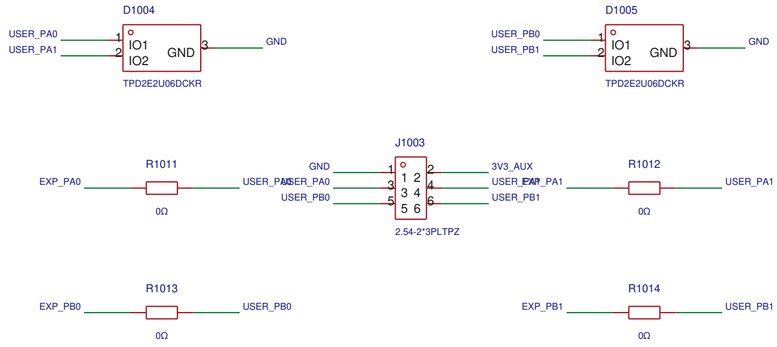
1 GND / 2 3V3_AUX / 3 SCL / 4 SDA
4.7k 上拉；外部重复上拉需合并计算
AUX 关断时停用总线，禁用内部上拉

5V_PORT 共享限流 / 故障锁存



NRST 低时关断；故障锁存后复位核心板重试
45.3k ILIM；外部连续预算 400mA
PORT_FAULT_N → PE6，并引至扩展板

自定义 GPIO / 2x3 直立排针 / 2.54mm

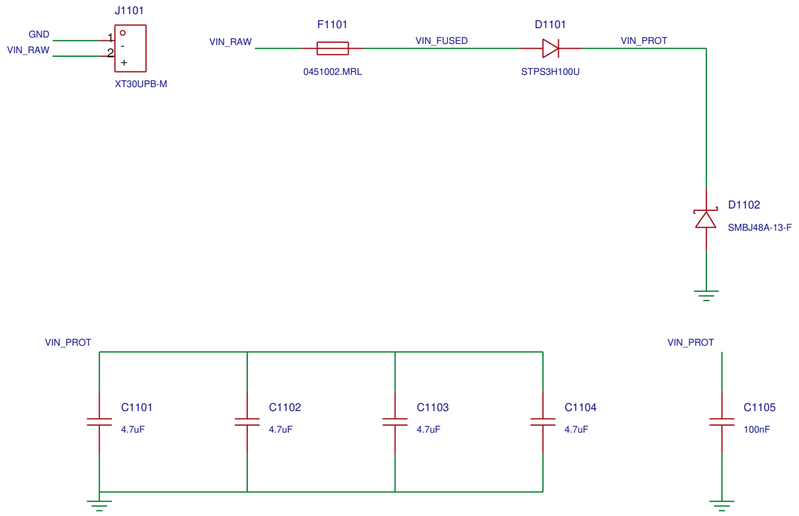


1 GND / 2 3V3_AUX / 3 PA0 / 4 PA1 / 5 PB0 / 6 PB1
四路均可配置 GPIO；ADC/PWM 复用由固件决定
0Ω 用于隔离/测量；不得接入 5V、12V、24V 或 48V

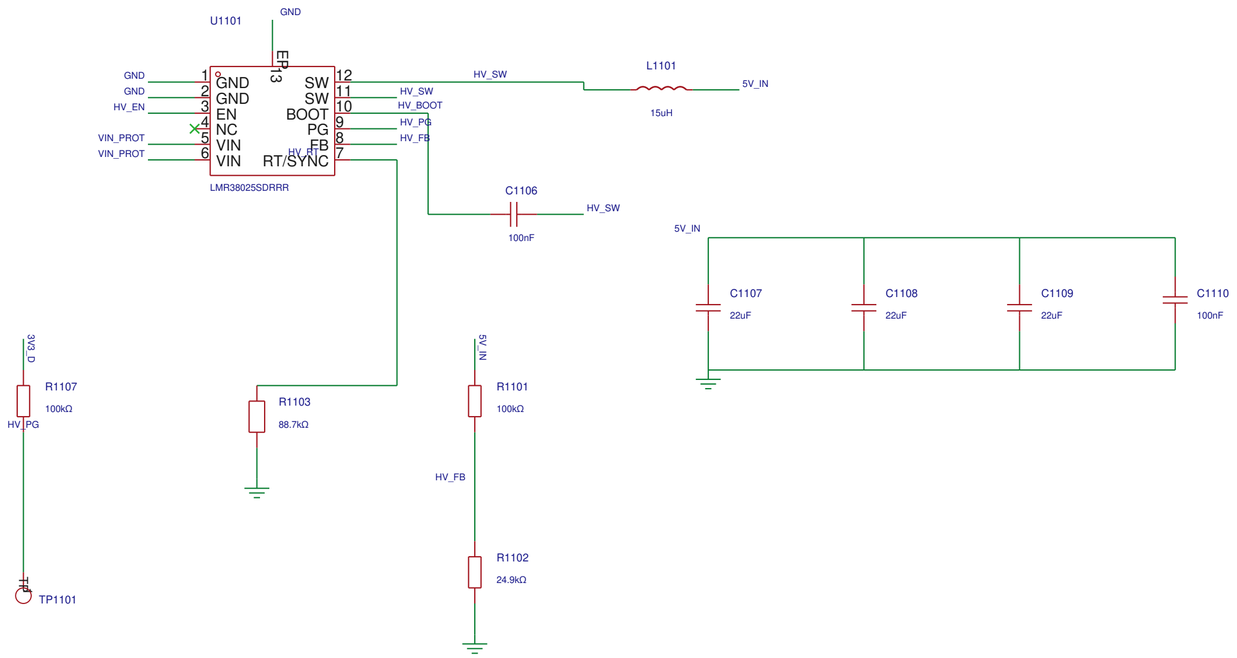
Native-source geometry preview / not an editor-rendered or fabrication drawing

C10_XT30宽压输入与5V电源

XT30 / 12–48V 稳压电源 / 5V 2A 连续预算

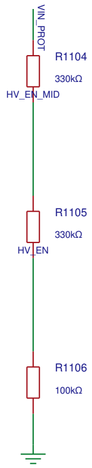


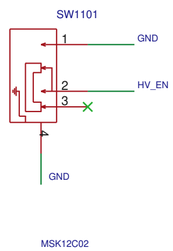
100V 输入电容：48V 偏压后总有效容量需 ≥4.7μF
48V TVS：钳位与插拔振铃必须实测；输入未做主动浪涌限流



LMR38025 / 300kHz / 15μH / 5.016V
输出电容：3×22μF 16V；额定 2.5A，热预算按 2A
HV_SW / BOOT 高频回路紧凑放置，远离 IMU 热岛

低压差启动 / EN 开关





330k + 330k / 100k：EN 欠压门限约 9.5V
SW1101 仅关闭降压输出，XT30 输入端仍带电
保险丝 2A / 125VDC；反接保护为 100V 串联肖特基
输入最大 48V，不能直接接入满电超过 48V 的电池

Native-source geometry preview / not an editor-rendered or fabrication drawing